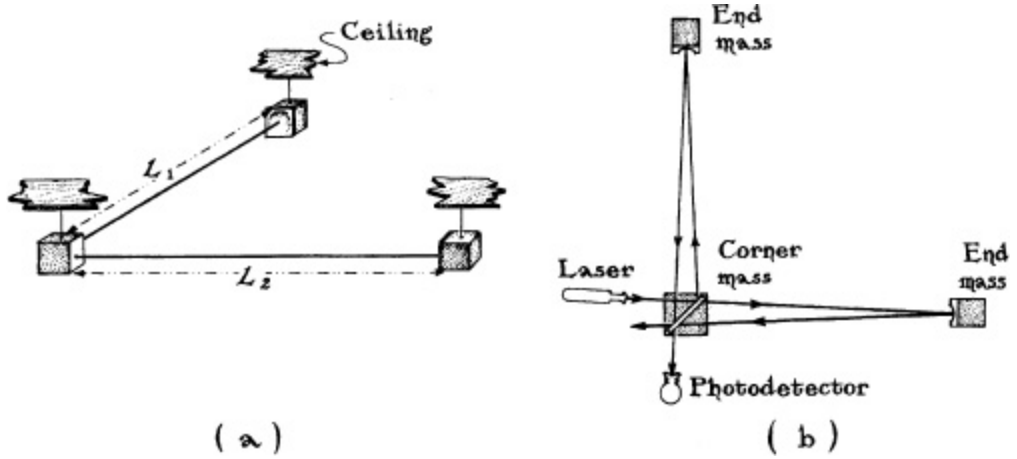
corner and ends of an "L" (Figure 10.6a). When the first crest of a gravitational wave enters the laboratory from overhead or underfoot, its tidal forces should stretch the masses apart along one arm of the "L" while squeezing them together along the other arm. The result will be an increase in the length L_1 of the first arm (that is, in the distance between the arm's two masses) and a decrease in the length L_2 of the second arm, When the wave's first crest has passed and its first trough arrives, the directions of stretch and squeeze will be changed: L_1 will decrease and L_2 will increase. By monitoring the arm-length difference, $L_1 - L_2$, one can seek gravitational waves.



10.6 A laser interferometric gravitational-wave detector. This instrument is very similar to the one used by Michelson and Morley in 1887 to search for motion of the Earth through the aether (Chapter 1). See the text for a detailed explanation.

The difference $L_1 - L_2$ is monitored using *interferometry* (Figure 10.6b and Box 10.3). A laser beam shines onto a *beam splitter* that rides on the corner mass. The beam splitter reflects half of the beam and transmits half, and thereby splits the beam in two. The two beams go down the two arms of the interferometer and bounce off mirrors that ride on the arms' end masses, and then return to the beam splitter. The splitter half-transmits and half-reflects each of the beams, so part of each beam's light is combined with part from the other and goes back toward the laser, and the other parts of the two beams are combined and go toward the photodetector. When no gravitational wave is present, the contributions from the two arms interfere in such a way (Box 10.3) that all the net light goes back toward the laser and none toward the photodetector. If a gravitational wave slightly changes